ANSWER KEY & MARKING SCHEME SET A

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** garden pea	Pisum sativum
2	**(d)** father	X or Y sperm
3	**(d)** all tall	Tall is dominant
4	**(a)** Correct	Genotypic is 1 : 2 : 1
5	**(b)** Student A is correct	Female XX
6	**(b)** genes	On DNA
7	**(d)** the temperature at which eggs are incubated	Environmental
8	**(c)** 9 : 3 : 3 : 1	Independent assortment
9	**(a)** dominant	Masks the other
10	**(a)** tt	Homozygous recessive

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The transmission of characters (traits) from parents to offspring.	2
12	TT : Tt : tt = 1 : 2 : 1	2
13	Many contrasting visible traits; self-pollinating; short life cycle; many seeds.	2
14	True — a girl gets X from the mother and X from the father; a boy gets Y from the father.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Genes carry information to make proteins (enzymes); e.g. an efficient enzyme makes more growth hormone → tall plant; less efficient → short plant	3
16	Mother: XX → all eggs X; father: XY → sperms X or Y; X sperm + X egg → girl (XX); Y sperm + X egg → boy (XY). 50% chance each.	3
17	F1: all round yellow (RrYy); F2: round yellow 9, round green 3, wrinkled yellow 3, wrinkled green 1 (9 : 3 : 3 : 1) — traits are inherited independently	1+2
	OR: Dominant: expressed in F1 (tall); recessive: hidden in F1 and reappears in F2 (short)	3
18	Each parent contributes one copy of each gene through gametes that carry half the chromosomes; the zygote gets one set from each parent	3
	OR: Tall parent Tt × short tt → Tt (tall) : tt (short) = 1 : 1	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) TT × tt → F1 Tt (all tall); F1 self-crossed: TT, Tt, Tt, tt → 3 tall : 1 short (ii) in a hybrid, only the dominant trait is expressed (iii) first and second filial generations	3+1+1
	OR: (i) RRYY × rryy → RrYy; F2: 9 RY : 3 Ry : 3 rY : 1 ry (ii) independent inheritance of traits (iii) round green and wrinkled yellow 3 : 3 (6/16)	3+1+1
20	(i) father's sperm (ii) XX × XY → XX (girl), XY (boy) (iii) 50% (iv) sex is a matter of chance; the mother is not responsible	1+2+1+1
	OR: (i) carrier of hereditary information (ii) section of DNA that codes for a protein (iii) gene → enzyme → amount of hormone → height (iv) on chromosomes in the nucleus	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 8: **Heredity** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 23	22 autosomes + 1 sex
2	**(b)** 75%	3/4
3	**(b)** TT	Homozygous
4	**(c)** Incorrect — they reappear in F2	Hidden only
5	**(a)** Student A is correct	Two different alleles
6	**(d)** 50%	X or Y sperm
7	**(a)** recessive	Yellow is dominant
8	**(d)** DNA	Hereditary material
9	**(d)** round yellow	Both dominant
10	**(d)** 1	rryy

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Alleles; they separate so each gamete gets only one.	2
12	Genotype: genetic make-up (TT/Tt); phenotype: observable trait (tall).	1+1
13	A cross considering only one pair of contrasting characters (e.g. tall vs short).	2
14	False — females have a perfect pair (XX); males have a mismatched pair (XY).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	F1: all violet (Vv); F2: VV, Vv, Vv, vv → 3 violet : 1 white	1+2
16	Autosomes: chromosomes not determining sex (22 pairs); sex chromosomes: determine sex (1 pair — XX or XY)	3
17	Round yellow × wrinkled green gives new combinations (round green, wrinkled yellow) in F2, showing the seed shape and colour genes are inherited independently	3
	OR: They are changes in non-reproductive (somatic) tissues and do not change the DNA of germ cells	3
18	Helps in survival under changing environments; basis for evolution; adaptation	3
	OR: Tall × short gave all tall F1 (short hidden), but short reappeared in F2 in 1/4, showing short is recessive and tall dominant	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) RrYy — round yellow (ii) RY, Ry, rY, ry (iii) 9/16	2+2+1
	OR: (i) law of dominance, law of segregation, law of independent assortment (ii) tall × short; round yellow × wrinkled green (iii) he used large numbers, mathematics and controlled crosses	2+2+1
20	(i) Pp (ii) Pp × Pp → PP, Pp, Pp, pp (iii) 1/4 (iv) purple	1+2+1+1
	OR: (i) by sex chromosomes; father's sperm decides (ii) XX × XY → XX, XY (iii) snails can change sex; in reptiles temperature of egg incubation	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.